

VEDANT NAIK

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EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

Expected May 2028

B.S. Mechanical Engineering

Relevant Coursework: Dynamics and Controls I, Electronics for Mechanical Engineers, Thermal Fluids Engineering I, Mechanics and Materials, Design and Manufacturing I, Numerical Computation for Engineers, Differential Equations, Classical Mechanics, Programming in Python. *In progress (Fall 2026): Thermal Fluids Engineering II, Dynamics and Controls II, Design and Manufacturing II.*

EXPERIENCE

MIT Bioinstrumentation Lab, Cambridge, MA

Dec 2025 – Present

Stop-Rotor VTOL UAV Program — PI: Prof. Ian Hunter; advised by Denzel Carter; funded by the Paul E. Gray (1954) UROP Fund

Control Systems Engineer

Jun 2026 – Present

- Redesigned the existing swashplate actuator configuration in SolidWorks to increase rigidity, running a motion study to verify actuator clearances across the full range of travel.
- Developed vehicle controller in **Simulink** implementing force-to-command, CCPM mixing, and servo-horn linearization.
- Validated main rotor and swashplate CCPM logic through **hardware-in-the-loop** testing on Pixhawk 6X flight hardware, running the controller against a Simulink plant model.

Aerodynamics Researcher

Dec 2025 – Present

- Led wind tunnel testing and data acquisition for the aerodynamic evaluation of a set of novel VTOL airfoils in MIT AeroAstro's 2x3 ft tunnel, supporting a **Journal of Aircraft research submission**.
- Developed wind tunnel test rig, and designed wake-rake and manometer measurement procedures to quantify profile drag in consultation with **Prof. Mark Drela** (author of XFOIL).

MIT Motorsports, Cambridge, MA

Sep 2024 – Dec 2025

Vehicle Performance Engineer & Driver

- Developed the team's first driver-in-the-loop simulation with active controls using IPG Automotive software, giving drivers a closed-loop environment to evaluate setup changes before track time.
- Directed a sub-team to quantify, model, and simulate tire behavior across temperature and wear states using tire temperature sensors, wear indicators, and vehicle telemetry, extracting **20% more use** from tires during testing.
- Drove the 2025 vehicle at FSAE Electric to a **9th place overall** finish, the team's best result since 2017.

Seat Lead

- designed, analyzed, manufactured, and integrated the team's *first* structurally cored carbon fiber driver seat, using NX and Teamcenter whilst working and compromising with four subsystems, reducing seat and firewall system mass by **4 kg**.
- Tested and validated laminate and core deflection and yield strengths on an Instron 3-point bending rig.

PROJECTS

- **Autonomous Line-Following Competition Robot** (2.678 — Arduino, C++) — Implemented and tuned a weighted-average PD controller over a reflectance sensor array paired with a time-based state machine optimized for speed. Placed **2nd in a class of 50+** students.
- **3D Printed Bridge — David Good / NJIT Competition** — Led a design team to **1st place**, setting the high school, collegiate, and **overall test-bench records** by holding over 500 lb on a bridge weighing under 500 g. Mentored the following year's team to a 3rd place finish.

SKILLS & CERTIFICATIONS

Controls & Simulation: MATLAB/Simulink, Stateflow, PX4/Pixhawk, hardware-in-the-loop testing, control law design

Programming: Python, C, C++, MATLAB, R, Arduino/embedded C

CAD & Analysis: Siemens NX, SolidWorks, Autodesk Inventor, Fusion 360

Experimental: Wind tunnel testing, wake rake and manometry, data acquisition, sensor calibration, Instron mechanical testing

Manufacturing: CNC and manual milling, laser cutting, waterjet, 3D printing, composites manufacturing (wet layup, pre-preg)

Certifications: OSHA 10-Hour General Industry

ACTIVITIES

MIT Varsity Track and Field — 110m Hurdles, 400m Hurdles